

SERIGNE SALIOU MBACKÉ MBAYE

Research Engineer | Artificial Intelligence, Remote Sensing & Smart Agriculture

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RESEARCH PROFILE

Software Engineer and Research Engineer with an engineering degree in Computer Science/Software Engineering and hands-on experience in applied AI/ML, geospatial technologies and satellite remote sensing for agriculture. Current work includes crop yield estimation, crop classification and phenological analysis using Google Earth Engine and machine-learning methods including Ridge Regression and XGBoost. Interested in doctoral research at the intersection of Artificial Intelligence, Computer Vision, Remote Sensing, Agricultural Robotics and Autonomous Systems, with applications to precision agriculture and African agricultural challenges.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Engineer — Tolbi Co., Senegal

2026 – Present

- Satellite remote sensing applied to agriculture, including crop yield estimation, crop detection/classification and phenological analysis.
- Uses Google Earth Engine for geospatial/remote-sensing workflows and machine-learning methods including Ridge Regression for yield estimation and XGBoost for crop classification.
- Research interests evolving toward AI-enabled agricultural robotics, intelligent perception and autonomous systems.

Software / Application Development — Tolbi Co.

2023 – 2025

- Developed several software applications; experience spanning software engineering, backend/frontend development and deployment.

Artificial Intelligence & Machine Learning Program — Zone01

2023 – 2026

- Completed a project-based AI/ML curriculum covering data analysis, machine learning, neural networks/deep learning, NLP, image processing, model evaluation and dimensionality reduction.
- Completed applied projects including Kaggle Titanic, NLP Scraper, Emotions Detector, SP500 Strategies and Credit Scoring.
- Tools and methods include Python, NumPy, Pandas, scikit-learn, Keras/TensorFlow concepts, NLTK/SpaCy, feature engineering, cross-validation, Random Forest, XGBoost and neural networks.

Software Engineering Projects — Various organizations / projects

2019 – 2022

- 2019: Six-month final internship at Washi SARL.
- 2020–2022: Built programming-learning, local-media and NoCode web/back-office applications, culminating in an interactive code-learning platform.

EDUCATION

Diplôme d'Ingénieur de Conception (DIC) in Software Engineering — École

2022

Supérieure Polytechnique, Université Cheikh Anta Diop de Dakar

- M2-level final year in Computer Science/Software Engineering. Final professional internship average: 17.08/20.
- Selected coursework: Deep Learning 18/20; Distributed Systems Architecture 16.67/20; Data Warehousing 15.5/20; Blockchain Technology 14/20; Information Systems Security 12/20.
- Final thesis/project: analysis, design and implementation of an interactive programming-course sharing platform with screen recording, interactive video playback and an in-browser programming sandbox.

Diplôme Supérieur de Technologie (DST) in Computer Science —

2017

Baccalauréat —

2017

SELECTED AI / ML PROJECT EXPERIENCE

- NLP Scraper: news-intelligence project involving NLP, text preprocessing, embeddings, sentiment analysis, named-entity recognition and classification.
- Emotions Detector: facial-emotion detection from webcam streams using CNN/deep-learning concepts, image processing, transfer learning and real-time video processing.
- SP500 Strategies: machine-learning financial strategy with feature engineering, model training, backtesting and evaluation.

- Credit Scoring: default-risk prediction with interpretable machine-learning models, feature engineering, evaluation and hyperparameter tuning.
- Kaggle Titanic: predictive modeling, data analysis, feature engineering, classification, cross-validation and model evaluation.

TECHNICAL SKILLS

AI / Machine Learning: Python, NumPy, Pandas, scikit-learn, PyTorch, Jupyter, Google Colab, Vertex AI; Ridge Regression, XGBoost, neural networks, deep learning, NLP and computer-vision foundations

Remote Sensing / GIS: Google Earth Engine, QGIS, PostGIS, Leaflet.js, Mapbox; agricultural satellite remote sensing

Software Engineering: TypeScript, Node.js/NestJS, Express.js, Golang/Gorilla, Python/Django, Java/Spring Boot, PHP/Laravel, C#/.NET

Front-end / Mobile: Vue.js/Nuxt, React/Next.js, Flutter, UI/UX

Deployment: GitHub, Fly.io, Railway, Vercel, Netlify, VPS deployment

RESEARCH INTERESTS

Artificial Intelligence • Agricultural Robotics • Computer Vision • Remote Sensing • Geospatial AI • Autonomous Systems • Precision Agriculture • Smart Agriculture

LANGUAGES

French: Fluent | **English:** Intermediate

ADDITIONAL INFORMATION

Nationality: Senegalese | Target: PhD admission, 2027 intake | Funding target: Chinese Government Scholarship (CSC)